Santiago López Begines, PhD

Bioinformatics Scientist | Proteomics Data Analyst | Senior Researcher

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SUMMARY

Bioinformatics Scientist with 15+ years in biomedical research, combining extensive wet-lab experience in **affinity-based protein techniques (co-immunoprecipitation, pull-downs), proteomics, and metabolomics** with 4–5 years of computational specialization in **machine learning, statistical modeling, and multi-omics data analysis**. Proficient in **Python and R** for building automated proteomics pipelines (MaxQuant output processing, differential expression), biomarker discovery, and data visualization on large-scale omics datasets. Experienced in **SQL, Git, REST APIs, and cloud-based platforms**. Proven ability to bridge wet-lab biology and computational analysis, with strong cross-functional collaboration and scientific communication skills.

SKILLS

Programming, Tools & Data

- **Languages:** Python (NumPy, Pandas, scikit-learn, TensorFlow, XGBoost, LightGBM), R (tidyverse, tidymodels, caret, H2O, Seurat, Shiny, ggplot2), SQL
- **Data & ML:** Predictive modeling, classification, time-series analysis, NLP/sentiment analysis, statistical testing, data visualization (matplotlib, seaborn, ggplot2)
- **Bioinformatics:** Multi-omics integration (proteomics, transcriptomics, metabolomics), MaxQuant/label-free quantification, protein identification, biomarker discovery, automated analysis pipelines
- **Infrastructure:** Git/GitHub, REST APIs, JSON, YAML, cloud platforms, Jupyter, Linux/Ubuntu
- **Visualization & Reporting:** R Shiny dashboards, ggplot2, matplotlib, seaborn, scientific documentation

Biochemistry & Molecular Biology

- **Protein Biochemistry:** Co-immunoprecipitation, affinity pull-down assays, protein–protein interaction studies (SNARE complex), protein expression/purification, Western blot, SDS-PAGE
- **Molecular Biology:** PCR/RT-PCR, DNA/RNA analysis, cloning, sequencing
- **Neuroscience:** Whole-cell patch-clamp electrophysiology, transgenic mouse models, surgical procedures (cranial window, AAV injections), confocal and electron microscopy
- **Engineering:** Microcontroller programming (Arduino, ESP8266/ESP32), electronic circuit design, 3D printing (Fusion360)

Soft Skills

- Cross-functional team leadership and coordination in multicultural, international environments
- Scientific project management: multiple parallel projects, SOP documentation, process optimization
- Technical communication: scientific reporting, documentation, and mentoring researchers and analysts
- Fluent English (verbal and written)

WORK EXPERIENCE

Senior Scientist – Data Analyst | University of Luxembourg (LCSB) | 2023 – 2025

- Developed and maintained proteomics data analysis pipelines (R/Python) processing Orbitrap/MaxQuant output for label-free quantification, differential protein expression, and biomarker discovery in neurodegenerative disease
- Applied machine learning models (classification, regression) to predict disease progression, improving feature selection through **statistical validation and cross-validation frameworks**
- **Reduced data cleaning time by 70%** by developing automated R scripts for omics data validation and transformation
- Conducted systematic literature reviews and database queries to extract and curate relevant biomedical data
- Created and maintained internal SOPs and documentation for reproducible analytical workflows
- Collaborated with cross-disciplinary international teams on EU-funded research projects

Senior Scientist | Instituto de Biomedicina de Sevilla (IBiS) | 2017 – 2023

- Analyzed proteomics and transcriptomics datasets to uncover molecular mechanisms in neurodegeneration; designed and executed co-immunoprecipitation and pull-down experiments for protein interaction studies
- Built R/Python scripts for statistical analysis, data visualization (ggplot2, matplotlib), and automated reporting
- **Developed automated electrophysiology analysis pipeline, saving up to 90% of processing time** compared to traditional manual analysis
- Performed whole-cell patch-clamp paired recordings in mouse brain slices, generating large electrophysiological datasets
- Supervised PhD candidates and contributed to internal training sessions on data analysis methods
- Managed data integrity across large-scale research projects, optimizing multi-team workflows between biology and informatics

Junior Scientist | University of Barcelona | 2009 – 2017

- Doctoral thesis focused on phototransduction retinal complex protein–protein interactions using co-immunoprecipitation and affinity pull-down assays, combined with quantitative electrophysiology and imaging analysis
- Managed transgenic mouse colony and experimental protocols, ensuring compliance with institutional standards
- Designed and optimized experimental workflows integrating wet-lab biochemistry with quantitative data analysis for publication in peer-reviewed journals

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PUBLICATIONS AND CONFERENCES

- **Co-authored 8+ peer-reviewed publications** in high-impact biomedical journals (ORCID: orcid.org/0000-0001-8809-8919)
- Presented research findings at international conferences (10+ oral and poster presentations)

EDUCATION

- **MSc in Big Data and Data Science** — UNED, Spain (2025)
- **PhD in Biomedicine** — University of Barcelona, Spain (2016)
- **MSc in Neurosciences** — University of Barcelona, Spain (2011)
- **BSc in Biochemistry** — University of Seville, Spain (2009)